UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Flowcharts and visuals

Procedures, decisions, concept maps and comparison tables.

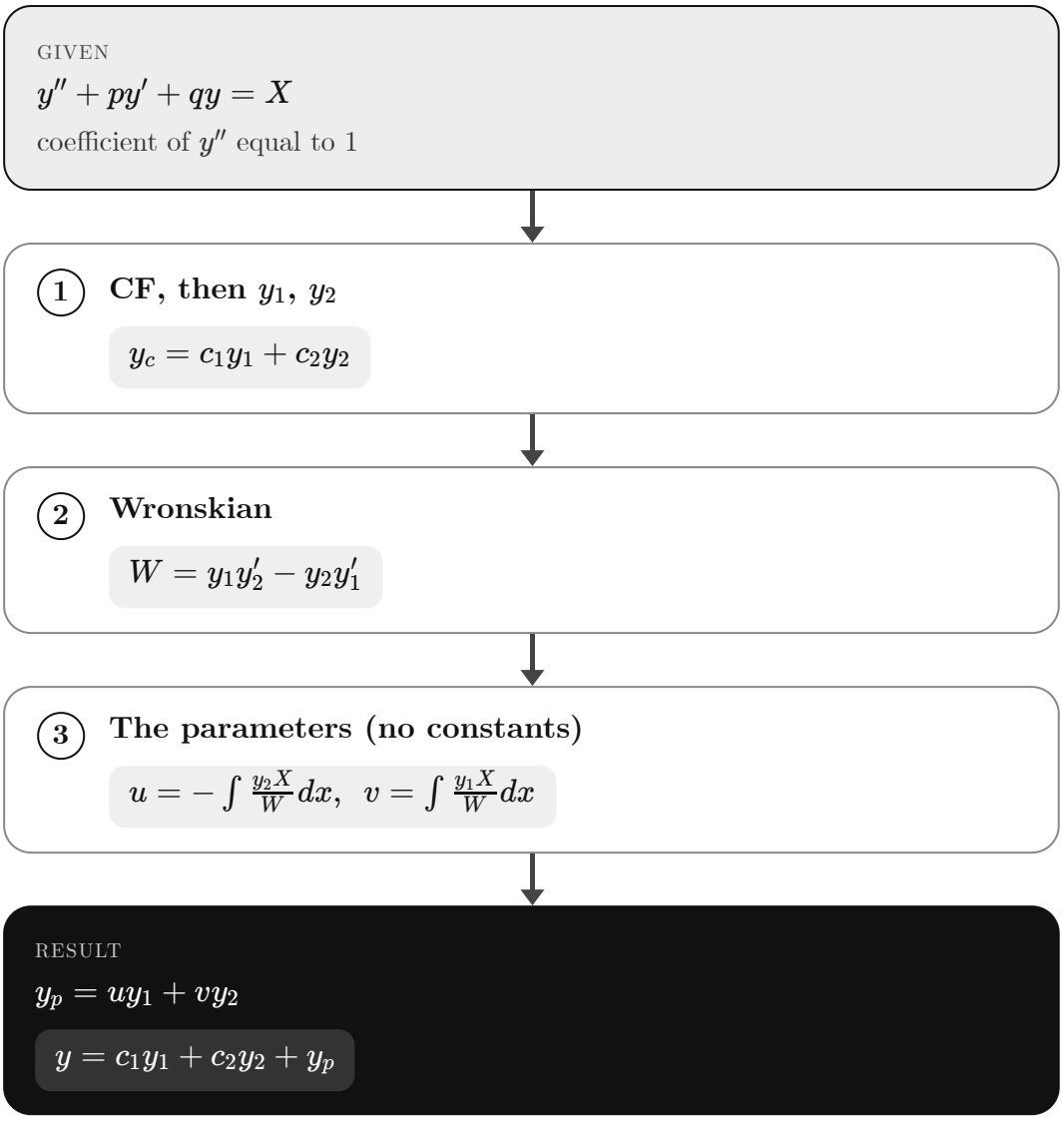
Class 011 (VoP) · Method of variation of parameters · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S011V·9	Unit 2 deck 011 “Variation of Parameters method” (SESSION 8), slide 9 (rendered .pptx)
T011V 00:03:26	AV-summary transcript “Variation of Parameters method” (14.9 min, listed first), segment start
T011Vb 00:16:53	AV-summary transcript “Variation of Parameters method_Continued” (24.5 min, second), segment start
CW U2 P25	Unit 1 & 2 class-work sheet, Unit 2 problem 25
ESA Jul-23 3c	Past paper: UE22MA141A July 2023, question 3.c (Jan-24 s1/s2 = UE23MA141A Jan 2024 sets 1/2)

Contents

1. Figure F1.1 · Variation of parameters in three steps (slides 011V·4–5)
2. Figure F1.2 · Before you start: the standard form (slide 011V·6)
3. Figure F1.3 · Which method for the PI?
4. Figure F2.1 · Slides 011V·14–17: $(D^2-1)y=\frac{2}{1+e^x}$
5. Figure F3.1 · ESA Jan-24 s1 3a: $(D^2-2D+2)y=e^x\tan x$
6. Figure F4.1 · ESA Jul-23 3c: $y''+6y'+9y=\frac{e^{-3x}}{x}$
7. Variation of parameters or the operator rules?
8. Every variation-of-parameters problem in this pack
9. Terms you may absorb into the CF

1 The method: $y_p = uy_1 + vy_2$ with the Wronskian

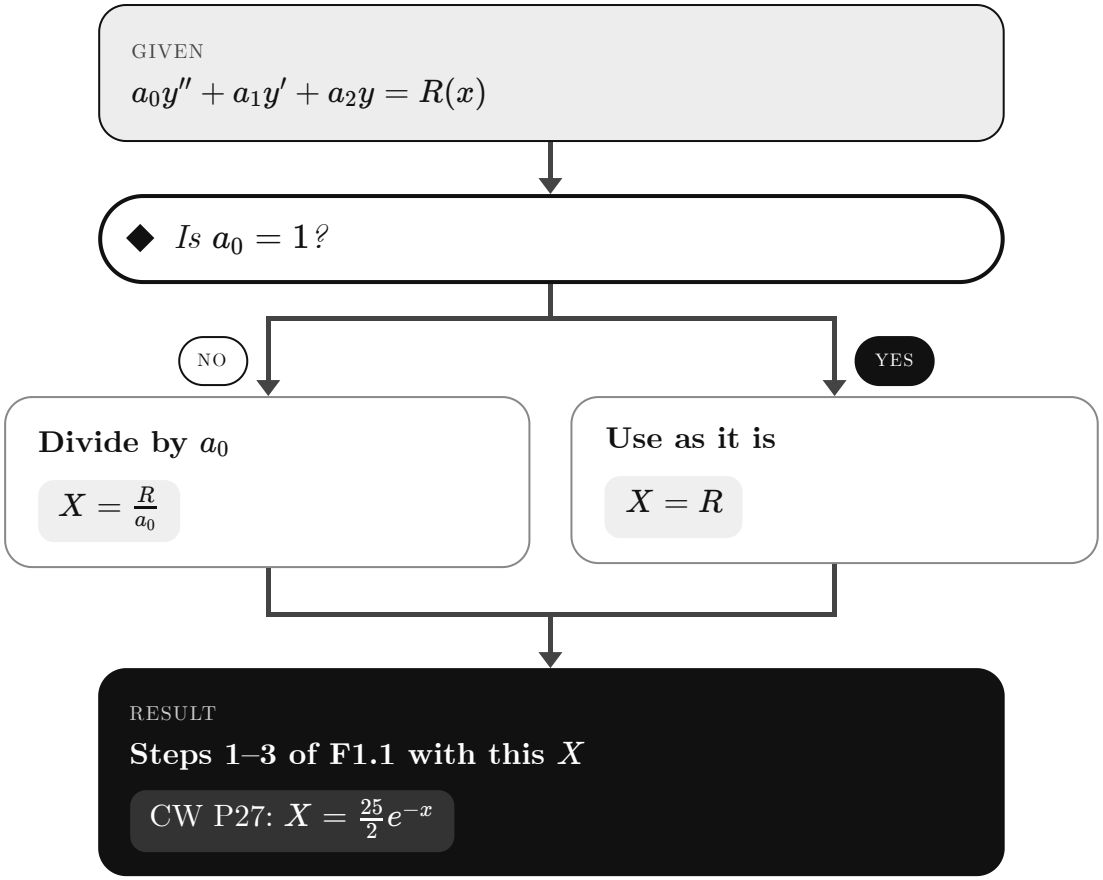


Remark. Terms of y_p that are multiples of y_1 or y_2 may be absorbed into c_1, c_2.

Exam trap. Minus sign on u; u has y_2 inside, v has y_1.

Figure F1.1. Variation of parameters in three steps (slides 011V·4–5) ESA ★★★

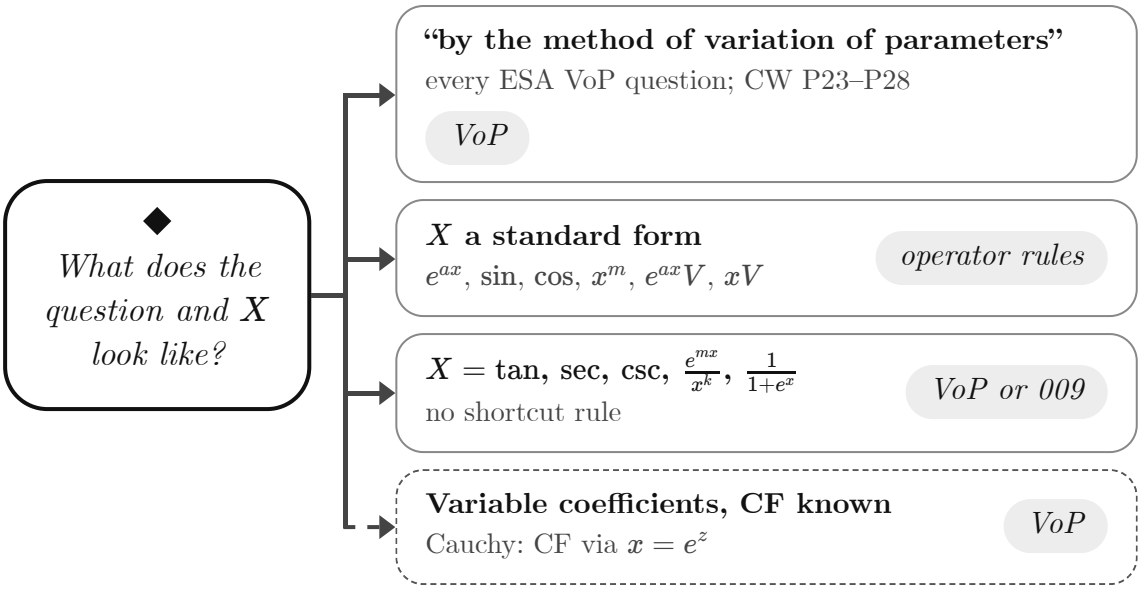
S011V·4 · S011V·5 · T011V 00:08:55 · T011Vb 00:08:20



Remark. The CF (AE roots) is the same either way; only X changes.

Figure F1.2. Before you start: the standard form (slide 011V·6) ESA ★★★

S011V·6 · T011V 00:06:06 · CW U2 P27



Remark. Class 009's operator method $\frac{1}{D-m}X = e^{mx}\int Xe^{-mx}dx$ is the alternative for non-standard X.

Figure F1.3. Which method for the PI? ESA ★★★

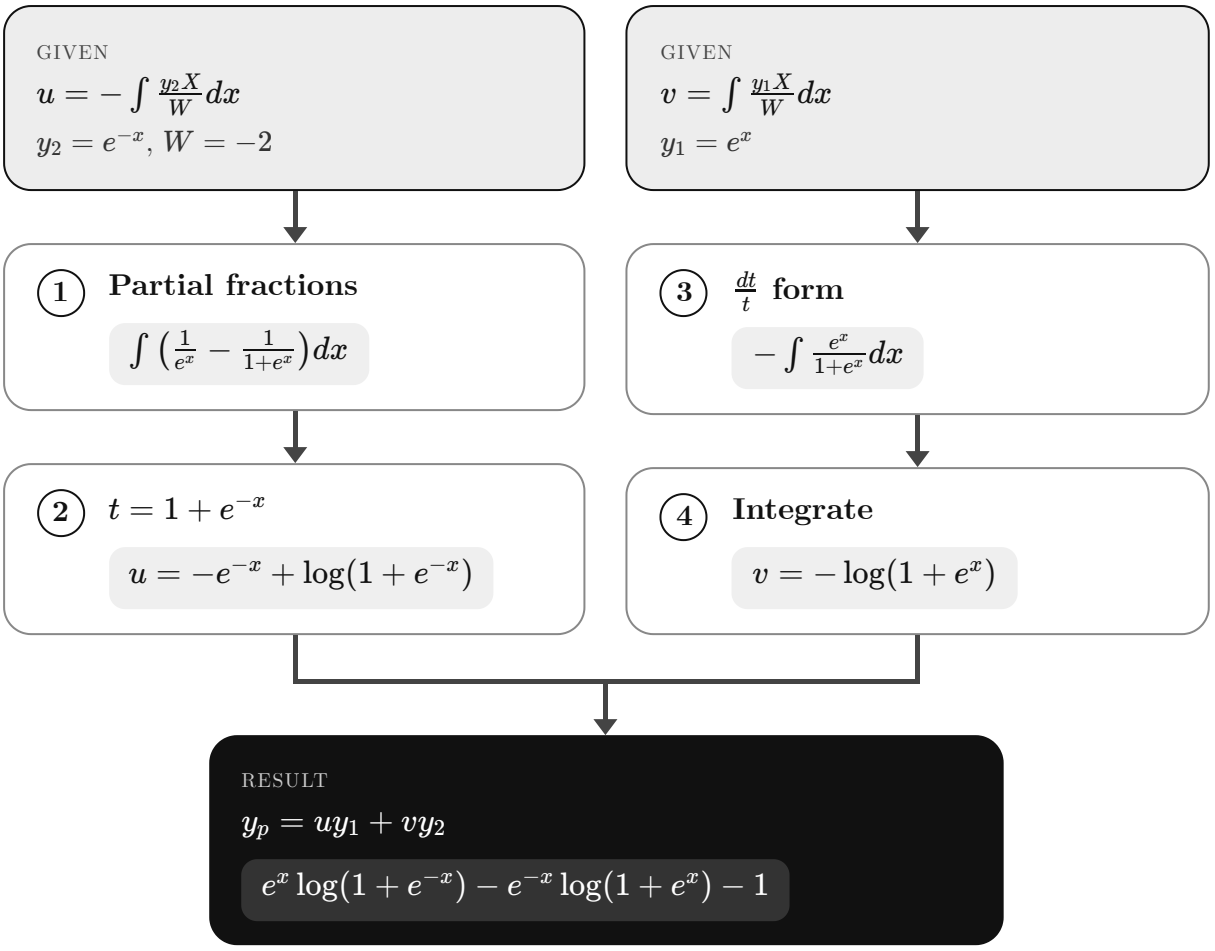
ADDED FOR CLARITY T011V 00:08:02 · T011V 00:08:19 · ESA Jul-23 3c · CW U2 P23

Variation of parameters or the operator rules?

	Operator rules (classes 003–006, 009)	Variation of parameters (this pack)
Needs the CF first?	no (PI alone)	yes: y_1, y_2 and W
Which X?	standard forms; 009 for any X via $\frac{1}{D-m}$	any X
Variable coefficients	only after a substitution	yes, if the CF is known
Main difficulty	choosing the rule; failure cases	the integrals
Use when	question says "solve"	question says "by the method of variation of parameters"

ADDED FOR CLARITY T011V 00:06:31 · T011V 00:08:02 · S011V·6

2 Slide and lecture examples: xe^{-x}, $\tan 2x$, $\frac{2}{1+e^x}$, a Cauchy equation

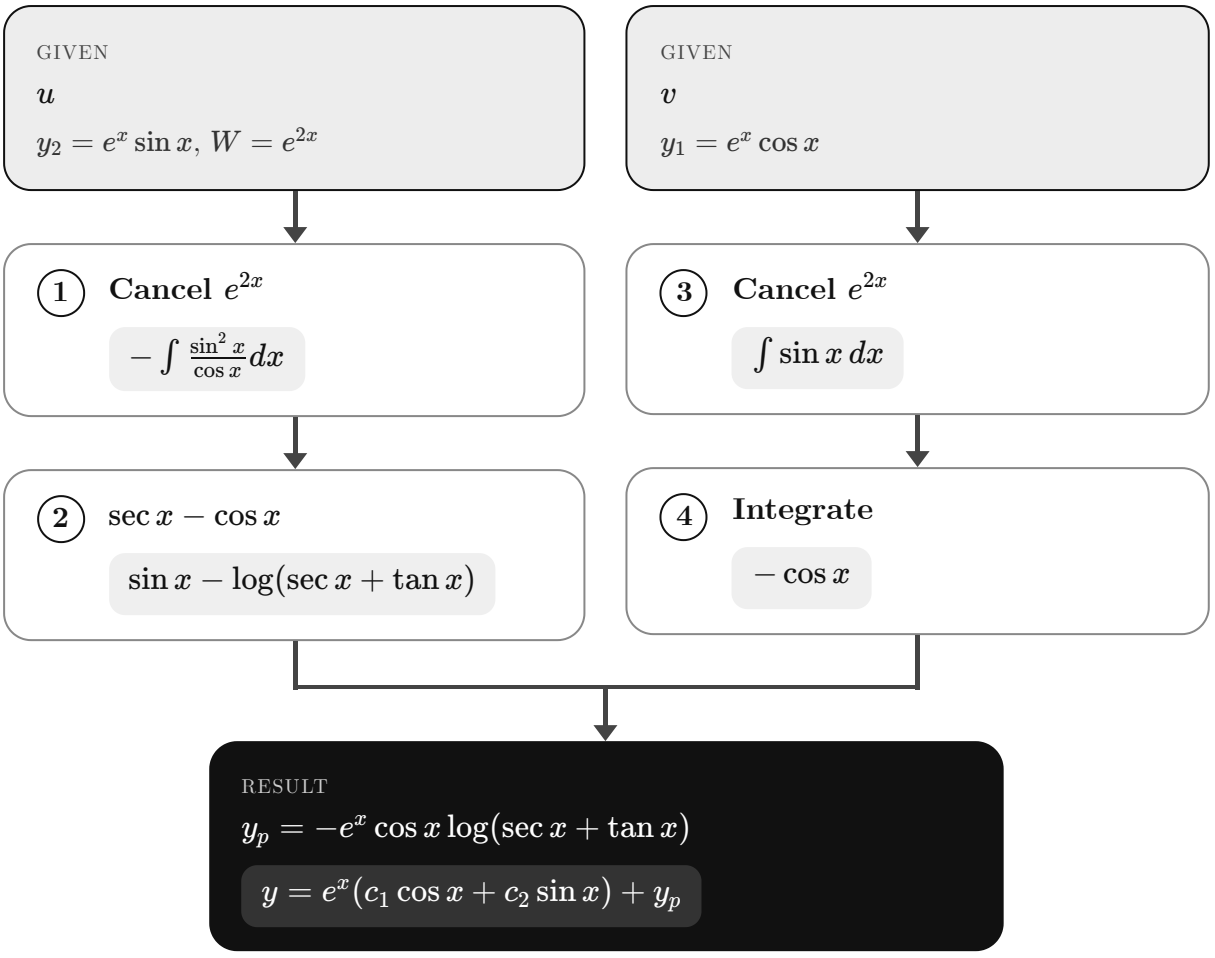


Remark. Worked in full as W2.3.

Figure F2.1. Slides 011V·14–17: $(D^2 - 1)y = \frac{2}{1+e^x}$ ESA ★★

S011V·14 · S011V·15 · S011V·16 · S011V·17

3 Trigonometric right-hand sides: $\sec ax$, $\csc x$, $e^{x}\tan x$, $e^{-x}\sec^{3} x$



Remark. The $e^{x}\sin x\cos x$ terms cancel. Worked in full as W3.3 (7 marks).

Figure F3.1. ESA Jan-24 s1 3a: $(D^2 - 2D + 2)y = e^{x}\tan x$ ESA ★★★

ESA Jan-24 s1 3a · CW U2 P25

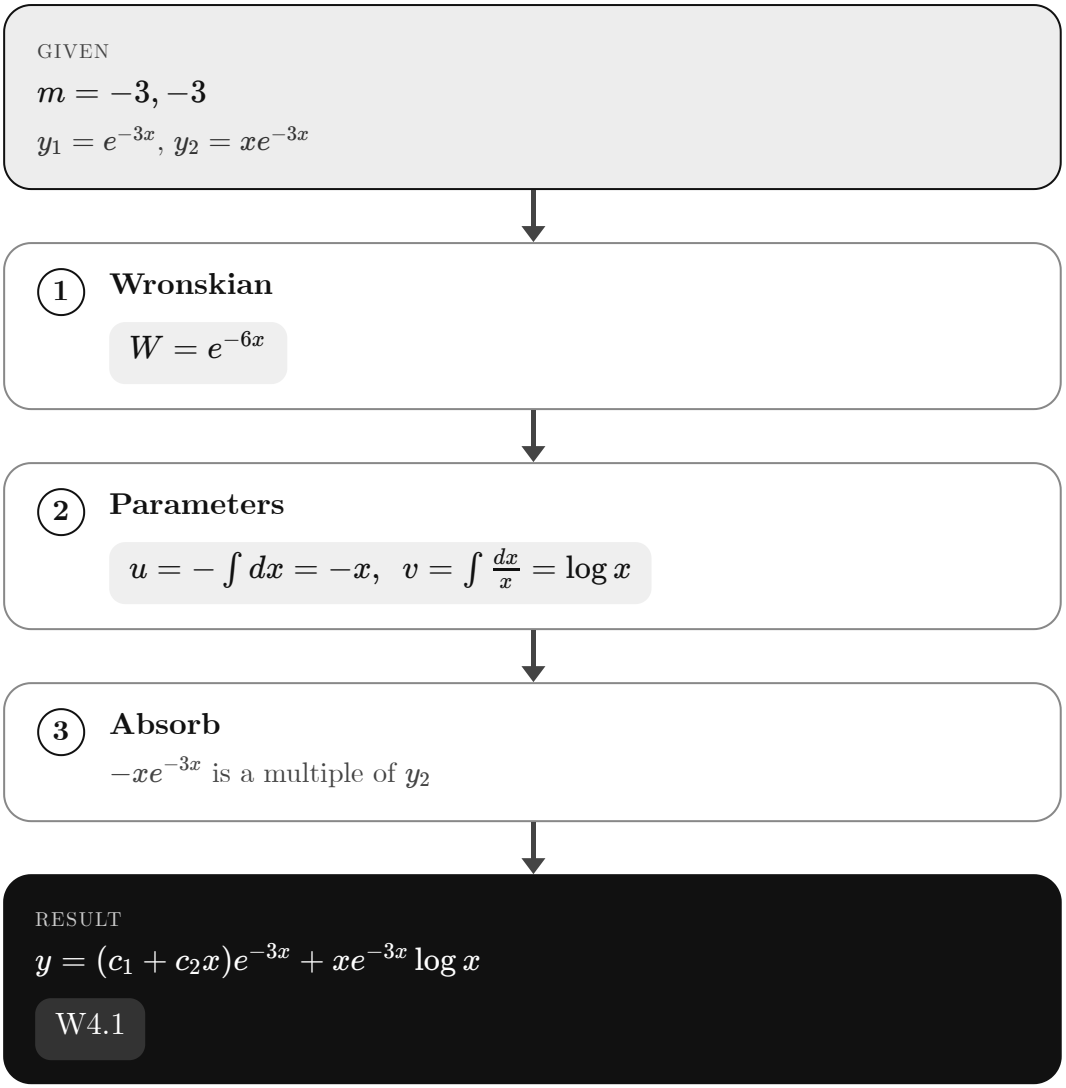
Every variation-of-parameters problem in this pack

Problem	y_1, y_2	W	u	v
xe^{-x}, $D^2 - 2D - 3$ **(W2.1)**	e^{3x}, e^{-x}	$-4e^{2x}$	$-\frac{e^{-4x}}{16}(x + \frac{1}{4})$	$-\frac{x^2}{8}$
$\tan 2x$, $D^2 + 4$ **(W2.2)**	$\cos 2x$, $\sin 2x$	2	$\frac{1}{4}[\sin 2x - \log(\sec 2x + \tan 2x)]$	$-\frac{\cos 2x}{4}$
$\frac{2}{1+e^x}$, $D^2 - 1$ **(W2.3)**	e^{x}, e^{-x}	-2	$-e^{-x} + \log(1 + e^{-x})$	$-\log(1 + e^{x})$

Problem	$y_1,\ y_2$	W	u	v
$\sec ax$ **(W3.1)**	$\cos ax,\ \sin ax$	a	$\frac{1}{a^2}\log\cos ax$	$\frac{x}{a}$
$\csc x$ **(W3.2)**	$\cos x,\ \sin x$	1	$-x$	$\log\sin x$
$e^x\tan x$ **(W3.3, ESA)**	$e^x\cos x,\ e^x\sin x$	e^{2x}	$\sin x-\log(\sec x+\tan x)$	$-\cos x$
$e^{-x}\sec^3 x$ **(W3.4, ESA)**	$e^{-x}\cos x,$ $e^{-x}\sin x$	e^{-2x}	$-\frac{\sec^2 x}{2}$	$\tan x$
$\frac{e^{-3x}}{x}$ **(W4.1, ESA)**	$e^{-3x},\ xe^{-3x}$	e^{-6x}	$-x$	$\log x$
$\frac{e^x}{x}$ **(W4.2, ESA)**	$e^x,\ xe^x$	e^{2x}	$-x$	$\log x$
$\frac{e^{-x}}{x^2}$ **(W4.3)**	$e^{-x},\ xe^{-x}$	e^{-2x}	$-\log x$	$-\frac{1}{x}$
$\frac{25}{2}e^{-x},\ -3y$ **(W4.4)**	$e^{3x/2},\ e^{-x}$	$-\frac{5}{2}e^{x/2}$	$-2e^{-5x/2}$	$-5x$

S011V·7 · S011V·11 · S011V·14 · CW U2 P23 · CW U2 P24 · CW U2 P25 · CW U2 P26 · CW U2 P27 · ESA Jul-23 3c · ESA May-24 3c · ESA Jan-24 s2 3a

4 Repeated roots with $X = \frac{e^{mx}}{x^k}$, and the leading-coefficient trap



Remark. May-24 3c is the same with $m = 1$: $y = (c_1 + c_2x)e^x + xe^x \log x$.

Figure F4.1. ESA Jul-23 3c: $y'' + 6y' + 9y = \frac{e^{-3x}}{x}$ ESA ★★★

ESA Jul-23 3c · ESA Jun-24 3a

Terms you may absorb into the CF

Problem	Raw $uy_1 + vy_2$	Absorbed term	Final PI
W2.1	$-\frac{x^2e^{-x}}{8} - \frac{xe^{-x}}{16} - \frac{e^{-x}}{64}$	$-\frac{e^{-x}}{64}$ (into c_2)	$-\frac{x^2e^{-x}}{8} - \frac{xe^{-x}}{16}$
W4.1	$xe^{-3x}\log x - xe^{-3x}$	$-xe^{-3x}$	$xe^{-3x}\log x$
W4.2	$xe^{x}\log x - xe^{x}$	$-xe^{x}$	$xe^{x}\log x$
W4.3	$-e^{-x}\log x - e^{-x}$	$-e^{-x}$	$-e^{-x}\log x$
W4.4	$-2e^{-x} - 5xe^{-x}$	$-2e^{-x}$ (optional)	$-5xe^{-x}$
W3.4	$-\frac{e^{-x}\cos 2x}{2\cos x}$	$-\frac{1}{2}e^{-x}\cos x$	$\frac{1}{2}e^{-x}\sin x\tan x$
W2.3	$e^{x}\log(1+e^{-x}) - e^{-x}\log(1+e^{x}) - 1$	none (-1 is not in the CF)	unchanged

ADDED FOR CLARITY S011V·10 · ESA Jul-23 3c · ESA May-24 3c · CW U2 P24 · CW U2 P27 · ESA Jan-24 s2 3a